



Fort Street High School

2022

HSC TRIAL EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6 – 11)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

NAME: _____

TEACHER: _____

STUDENT NUMBER:

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Question	1-10	11	12	13	14	15	16	Total
Mark	/10	/16	/16	/15	/14	/14	/15	/100

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

1 What is the unit vector in the same direction as $\underline{a} = 2i + j - 3k$?

A. $\frac{-1}{14} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$

B. $\frac{-1}{14} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

C. $\frac{-1}{\sqrt{14}} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$

D. $\frac{1}{\sqrt{14}} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}$

2 Given that $z^n = a + bi$ and $|z| = 1$ where a and b are real, which of the following is equal to z^{-n} ?

A. $a + bi$

B. $a - bi$

C. $-a + bi$

D. $-a - bi$

3 Consider the statement below:

“If z is even, then x and y are either both even or both odd.”

Which of the following is equivalent to the above statement?

- A. If x and y are either both even or both odd, then z is even.
- B. If z is odd, then exactly one of x and y is even.
- C. If x and y are either both even or both odd, then z is odd.
- D. If exactly one of x and y is even, then z is odd.

4 Consider the vectors $\overrightarrow{OA} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\overrightarrow{OB} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ and $\overrightarrow{OC} = \begin{bmatrix} 3 \\ \mu \\ \lambda \end{bmatrix}$, where O is the origin.

What are the values of μ and λ if the points A , B and C are collinear?

- A. $\mu = 3$, $\lambda = 2$
- B. $\mu = -3$, $\lambda = 2$
- C. $\mu = 3$, $\lambda = -2$
- D. $\mu = -3$, $\lambda = -2$

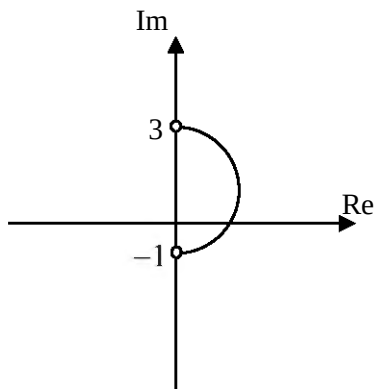
5 One of the roots of the quadratic equation $iz^2 + 3z + 3 - 11i = 0$ is $3 + 2i$.

What is the other root?

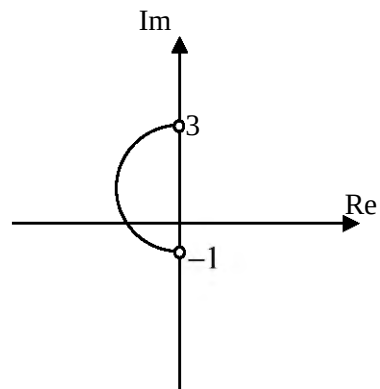
- A. $-2i$
- B. $-3 + i$
- C. $3 - 2i$
- D. $-5 - 3i$

6 Which of the following is the locus of z such that $\arg(z + 3i) - \arg(z - i) = \frac{\pi}{2}$?

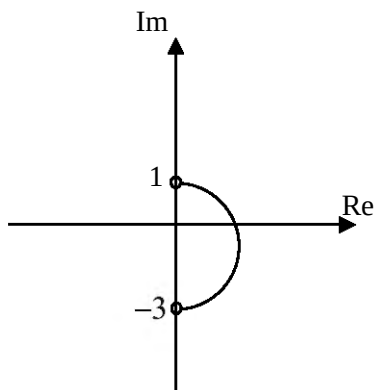
A.



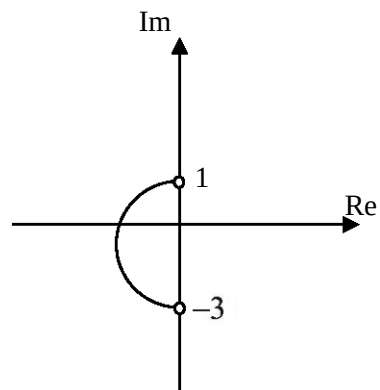
B.



C.



D.



7 Which of the following definite integrals has a positive value?

A. $\int_{-\frac{1}{2}}^{\frac{1}{2}} x^3(1 + \cos x) dx$

B. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\sin^{-1} x}{1 + x^4} \right) dx$

C. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\cos^{-1} x}{e^{-x^2}} \right) dx$

D. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\tan^{-1} x}{e^x + e^{-x}} \right) dx$

- 8 Given that z is a complex number such that $\operatorname{Re}(z) \neq 0$, which of the following is

equivalent to $\frac{4z\bar{z}}{(z+\bar{z})^2}$?

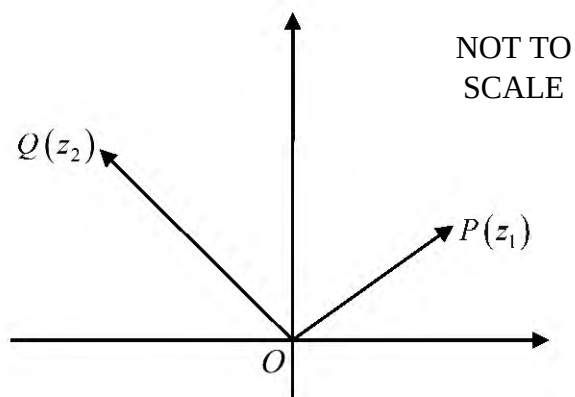
- A. $1 + \left(\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)} \right)^2$
- B. $4(\operatorname{Im}(z) \times \operatorname{Re}(z))$
- C. $4\left(1 + [\operatorname{Im}(z) + \operatorname{Re}(z)]^2\right)$
- D. $\frac{2 \times \operatorname{Im}(z)}{[\operatorname{Re}(z)]^2}$

- 9 Which of the following is the negation of the statement:

“ $\forall p \in P$ p is of the form $4m+1 \Rightarrow p$ can be written as a sum of two squares”?

- A. $\forall p \in P$, p is of the form $4m+1$ and p cannot be written as a sum of two squares.
- B. $\exists p \in P$, p is not of the form $4m+1$ and p can be written as a sum of two squares.
- C. $\forall p \in P$, p is not of the form $4m+1$ or p cannot be written as a sum of two squares.
- D. $\exists p \in P$, p is of the form $4m+1$ and p cannot be written as a sum of two squares.

- 10 In the diagram the vectors OP and OQ represent the complex numbers z_1 and z_2 respectively.



If $\frac{z_2}{z_1} = \sqrt{3}i$, what is the value of $\left| \frac{z_1 + z_2}{z_2 - z_1} \right|$?

- A. 1
- B. $\sqrt{3}$
- C. $\frac{1}{\sqrt{3}}$
- D. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use a SEPARATE writing booklet

- (a) If $z = -\sqrt{3} + i$ and $w = 1 + i$:
- (i) Find $\text{Arg}(zw)$ in terms of π . 2
- (ii) Write $\frac{z}{w}$ in the form $x + iy$, where x and y are real numbers. 2
- (b) Sketch the region on the Argand diagram where: 2
- $$\arg(z + 1 - 2i) = \arg(z - 3 + i).$$
- (c) Find the vector equation for the sphere $x^2 + 4x + y^2 - 6y + z^2 = 12$. 2
- (d) Point A has position vector $8\hat{i} + 13\hat{j} - 2\hat{k}$ the point B has position vector $10\hat{i} + 14\hat{j} - 4\hat{k}$ and the point C has position vector $9\hat{i} + 9\hat{j} + 6\hat{k}$.
- (i) Find $\overrightarrow{AB}$. 1
- (ii) Find $|\overrightarrow{CB}|$. 2
- (iii) What is the size of the acute angle between $\overrightarrow{AB}$ and $\overrightarrow{CB}$? 3
Answer correct to the nearest degree.
- (e) Find $\int \frac{1}{\sqrt{5 + 4x - x^2}} dx$. 2

End of Question 11

Question 12 (16 marks) Use a SEPARATE writing booklet

(a) (i) Find A and B such that $\frac{15}{x^2 - 2x - 15} = \frac{A}{x + 3} + \frac{B}{x - 5}$. 2

(ii) Hence or otherwise find $\int \frac{x^2 - 2x}{x^2 - 2x - 15} dx$. 2

(b) Evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \tan^2 x \, dx$. 3

(c) Use proof by contrapositive to prove that: 3

If $a^2 - 2a + 7$ is even then a is odd.

(d) Solve $|e^{i\theta} + \sqrt{2}| = 1$. 3

(e) Use the substitution $x = 2 \sin \theta$ to find $\int \frac{dx}{(4 - x^2)^{\frac{3}{2}}}$. 3

End of Question 12

Question 13 (14 marks) Use a SEPARATE writing booklet

(a) Find $\int \cos^{-1} x \, dx$ 3

(b) Use the substitution $t = \tan \frac{x}{2}$ or otherwise to evaluate: 3

$$\int_0^{\frac{\pi}{2}} \frac{1}{3 \sin x - 4 \cos x + 5} dx.$$

(c) The line l_1 passes through the points $A(2, -1, 4)$ and $B(4, -3, 2)$.

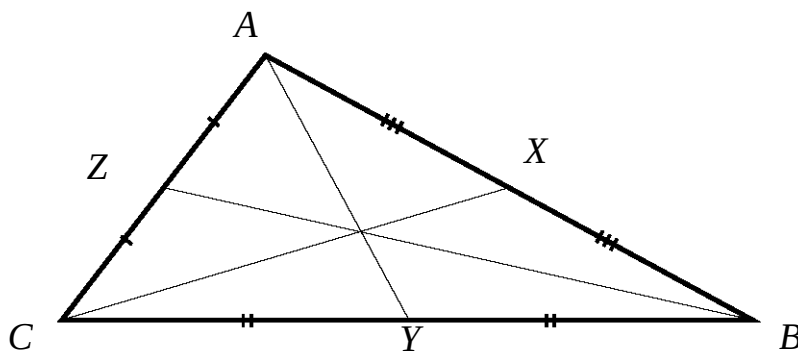
(i) Show that a vector equation of line l_1 is $\underline{r} = \lambda \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$. 2

(ii) The equation of line l_2 is given by $\frac{x+3}{k} = \frac{4-y}{6} = z-1$. 3

Find the value of k for which l_1 and l_2 intersect.

(d) The diagram shows a triangle ABC .

The points X , Y and Z bisect the intervals AB , BC and CA respectively.



Show that $\overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} = \mathbf{0}$

3

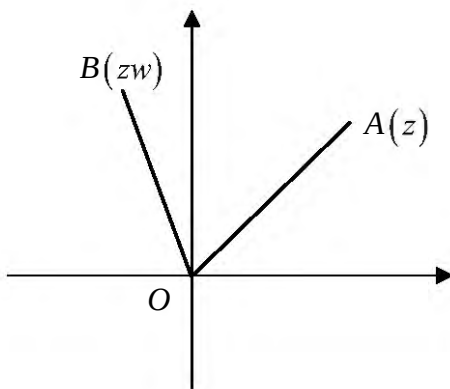
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) If $x \leq 1$ and $y \geq 1$ show that $x + y \geq 1 + xy$. 2

(b) Simplify fully $\frac{1}{1+\omega} + \frac{1}{1+\omega^2}$, given that ω is a non-real cube root of unity. 2

(c) As shown on the Argand diagram below, the complex numbers z and zw are represented by the points A and B respectively.



Given $z = re^{i\theta}$ and $w = e^{i\frac{\pi}{3}}$, where $r > 0$,

(i) Explain why OAB is an equilateral triangle. 2

(ii) Write the complex number $z - zw$ in exponential form in terms of r and θ . 2

(d) (i) Use a suitable substitution to show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 1

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x + \sin x} dx$. 3

(e) A recurrence relation is defined by $u_{n+1} = \frac{3u_n - 1}{4u_n - 1}$ and $u_1 = 1$. 3

Use Mathematical Induction to prove that $u_n = \frac{n}{2n-1}$ for $n \geq 1$.

End of Question 14

Question 15 (14 marks) Use a SEPARATE writing booklet

(a) (i) Show that $a^2 + 9b^2 \geq 6ab$, where a and b are real numbers. **1**

(ii) Hence, or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$. **2**

(iii) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$. **2**

(b) Consider the integral $I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} dx$. **4**

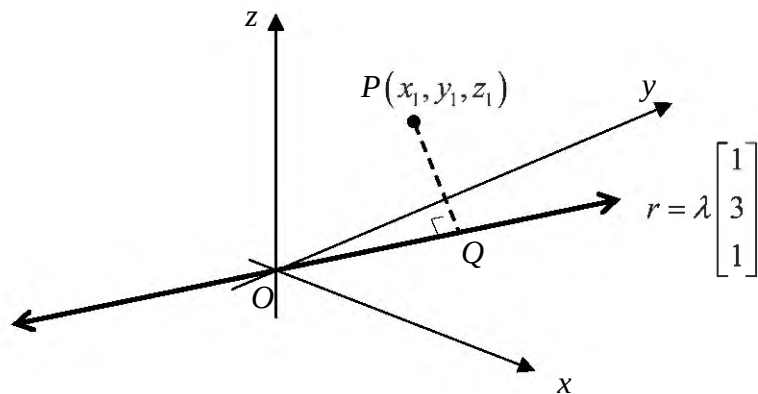
Use integration by parts to show that $I_n = \frac{2n}{2n+1} I_{n-1}$.

Question 15 Continues on Page 13

- (c) The diagram below shows the line with vector equation $\underline{r} = \lambda \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

The point $P(x_1, y_1, z_1)$ is any point in the three-dimensional space.

Q is the point on $\underline{r}$ such that PQ is a minimum.



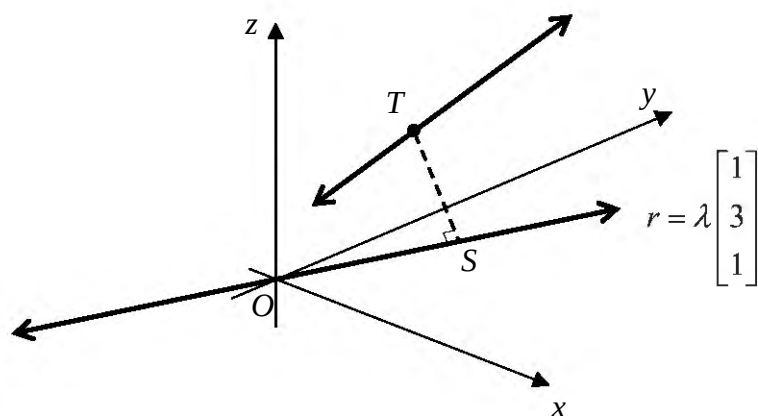
- (i) Using vector projection methods show that:

2

$$\overrightarrow{OQ} = \frac{x_1 + 3y_1 + z_1}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

It is given that line $\underline{c} = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$ does not intersect $\underline{r}$.

Point T is on the line $\underline{c}$ and S is a point on $\underline{r}$ such that TS is a minimum.



- (ii) By first expressing $\underline{c}$ in parametric form, find an expression for the vector $\overrightarrow{TS}$ in terms of t .

3

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Prove that: 2

$$\frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1} = i \tan \frac{\theta}{2}.$$

- (ii) Find the roots of the equation $w^5 = 1$. 2
Express your answers in the form $(\cos \theta + i \sin \theta)$ where $-\pi < \theta \leq \pi$.

- (iii) Using parts (i) and (ii) find the roots of the equation: 3

$$\left(\frac{2+z}{2-z} \right)^5 = 1$$

- (b) (i) Show that $\frac{1}{(k-1)k(k+1)} \geq \frac{1}{k^3}$ where k is a positive integer greater than 1. 2

- (ii) Given that: 3

$$S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} \quad \text{where } n \text{ is an integer and } n \geq 3.$$

Using part (i) or otherwise, prove that $S_n < \frac{1}{12}$.

- (c) Find $\int \frac{\ln x - 1}{[x + \ln x]^2} dx$. 3

End of paper

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

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What are the values of μ and λ if the points A , B and C are collinear?

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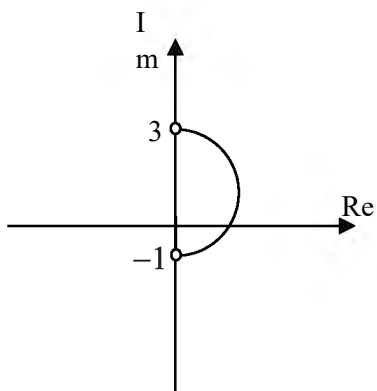
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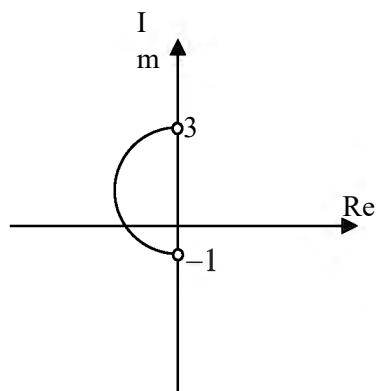
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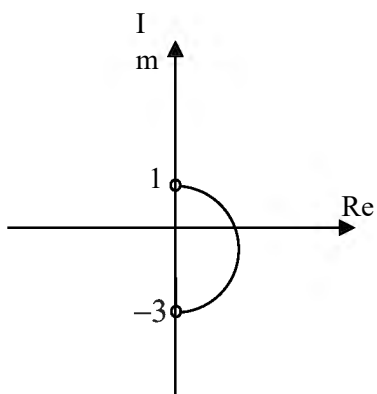
A.



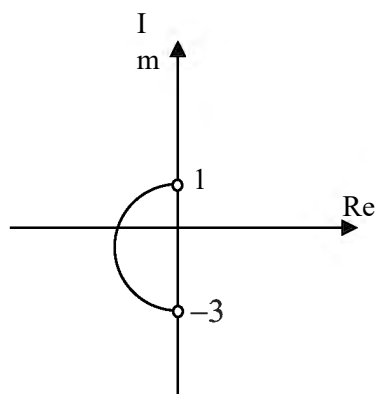
B.



C.



D.



7 Which of the following definite integrals has a positive value?

A. $\int_{-\frac{1}{2}}^{\frac{1}{2}} x^3 (1 + \cos x) dx$

B. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \left(\frac{\sin^{-1} x}{1 + x^4} \right) dx$

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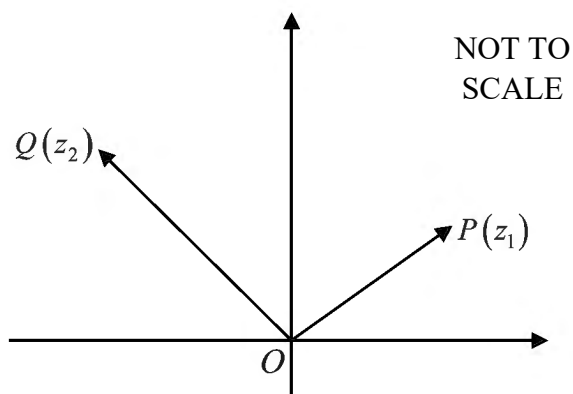
- ☒ A. $1 + \left(\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)} \right)^2$
- B. $4(\operatorname{Im}(z) \times \operatorname{Re}(z))$
- C. $4\left(1 + [\operatorname{Im}(z) + \operatorname{Re}(z)]^2\right)$
- D. $\frac{2 \times \operatorname{Im}(z)}{[\operatorname{Re}(z)]^2}$

- 9 Which of the following is the negation of the statement:

“ $\forall p \in P$ p is of the form $4m+1 \Rightarrow p$ can be written as a sum of two squares”?

- A. $\forall p \in P$, p is of the form $4m+1$ and p cannot be written as a sum of two squares.
- B. $\exists p \in P$, p is not of the form $4m+1$ and p can be written as a sum of two squares.
- C. $\forall p \in P$, p is not of the form $4m+1$ or p cannot be written as a sum of two squares.
- ☒ D. $\exists p \in P$, p is of the form $4m+1$ and p cannot be written as a sum of two squares.

- 10 In the diagram the vectors OP and OQ represent the complex numbers z_1 and z_2 respectively.



If $\frac{z_2}{z_1} = \sqrt{3}i$, what is the value of $\left| \frac{z_1 + z_2}{z_2 - z_1} \right|$?

- A. 1
- B. $\sqrt{3}$
- C. $\frac{1}{\sqrt{3}}$
- D. $\frac{\sqrt{3}+1}{\sqrt{3}-1}$

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (16 marks) Use a SEPARATE writing booklet

(a) If $z = -\sqrt{3} + i$ and $w = 1 + i$:

$$\arg z = \frac{5\pi}{6} \quad \text{and} \quad \arg w = \frac{\pi}{4}$$

$$\arg(zw) = \arg z + \arg w$$

$$= \frac{5\pi}{6} + \frac{\pi}{4}$$

$$= \frac{13\pi}{12}$$

$$\arg(zw) = -\frac{11\pi}{12}$$

(i) Find $\arg(zw)$ in terms of π .

2

$$\frac{w}{z} = \frac{1+i}{-\sqrt{3}+i}$$

$$= \frac{1+i}{-\sqrt{3}+i} \times \frac{1-i}{1-i}$$

$$= \frac{1+i}{-\sqrt{3}+i+i\sqrt{3}+1}$$

$$= \frac{\sqrt{2}}{1-\sqrt{3}+i+\sqrt{3}+1}$$

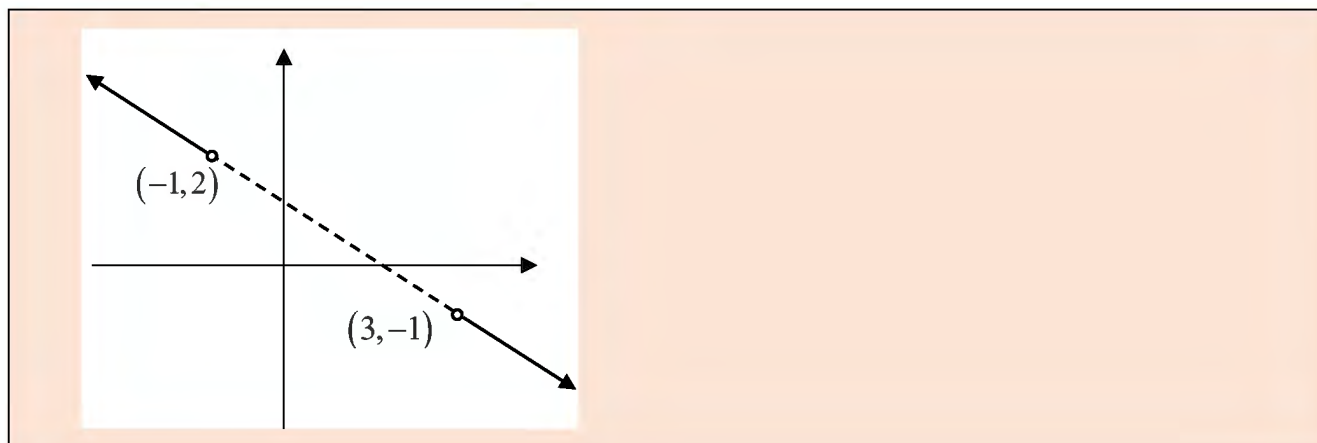
$$= \frac{\sqrt{2}}{1-\sqrt{3}+i+\sqrt{3}+1}$$

(ii) Write $\frac{w}{z}$ in the form $x + iy$, where x and y are real numbers.

2

(b) Sketch the region on the Argand diagram where:

2



$$\arg(z + 1 - 2i) = \arg(z - 3 + i).$$

$$(x^2 + 4x + 4) + (y^2 - 6y + 9) + z^2 = 12 + 4 + 9$$

$$(x + 2)^2 + (y - 3)^2 + z^2 = 25$$

Centre is $(-2, 3, 0)$ Radius is 5.

Vector equation is $\left| \vec{r} - \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix} \right| = 5$

(c) Find the vector equation for the sphere $x^2 + 4x + y^2 - 6y + z^2 = 12$.

2

(d) Point A has position vector $8\hat{i} + 13\hat{j} - 2\hat{k}$ the point B has position vector $10\hat{i} + 14\hat{j} - 4\hat{k}$ and the point C has position vector $9\hat{i} + 9\hat{j} + 6\hat{k}$.

$$\overrightarrow{AB} = \begin{bmatrix} 10 \\ 14 \\ -4 \end{bmatrix} - \begin{bmatrix} 8 \\ 13 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

(i) Find $\overrightarrow{AB}$.

1

(ii) Find $|\overrightarrow{CB}|$.

2

$$|\overrightarrow{CB}| = \sqrt{(9-10)^2 + (9-14)^2 + (6+4)^2} = \sqrt{1+25+100} = \sqrt{126} = 3\sqrt{14}$$

(iii) What is the size of the acute angle between $\overrightarrow{AB}$ and $\overrightarrow{CB}$?

3

$$\begin{aligned} \overrightarrow{AB} &= \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} & \text{Part i)} & \overrightarrow{BC} = \begin{bmatrix} 10-9 \\ 14-9 \\ -4-(-6) \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ -10 \end{bmatrix} \\ \cos \theta &= \frac{\overrightarrow{AB} \cdot \overrightarrow{CB}}{|\overrightarrow{AB}| |\overrightarrow{CB}|} = \frac{\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 5 \\ -10 \end{bmatrix}}{\sqrt{2^2+1^2+(-2)^2} \times 3\sqrt{14}} = \frac{3 \times 3\sqrt{14}}{2+5+20} = \frac{\sqrt{14}}{3} \\ \theta &= \cos^{-1}\left(\frac{\sqrt{14}}{3}\right) \\ \theta &\approx 37^\circ \end{aligned}$$

Answer correct to the nearest degree.

(e) Find $\int \frac{1}{\sqrt{5+4x-x^2}} dx$.

2

$$\begin{aligned}
 \int \frac{1}{\sqrt{5+4x-x^2}} dx &= \int \frac{1}{\sqrt{5-(x^2-4x)}} dx \\
 &= \int \frac{1}{\sqrt{5+4-(x^2-4x+4)}} dx \\
 &= \int \frac{1}{\sqrt{3^2-(x-2)^2}} dx \\
 &= \sin^{-1}\left(\frac{x-2}{3}\right) + C
 \end{aligned}$$

End of Question 11

Question 12 (16 marks) Use a SEPARATE writing booklet

- (a) (i) Find A and B such that $\frac{15}{x^2 - 2x - 15} = \frac{A}{x + 3} + \frac{B}{x - 5}$.

2

$$\frac{15}{x^2 - 2x - 15} = \frac{A}{x + 3} + \frac{B}{x - 5}$$

$$15 = A(x - 5) + B(x + 3)$$

Let $x = 5$

$$15 = +B(5 + 3)$$

$$\frac{15}{8} = B$$

Let $x = -3$

$$15 = A(-3 - 5)$$

$$-\frac{15}{8} = A$$

$$\frac{15}{x^2 - 2x - 15} = -\frac{15}{8(x + 3)} + \frac{15}{8(x - 5)}$$

1 – correct answer for A

1 – correct answer for B

- (ii) Hence or otherwise find $\int \frac{x^2 - 2x}{x^2 - 2x - 15} dx$.

2

$$\begin{aligned} \int \frac{x^2 - 2x}{x^2 - 2x - 15} dx &= \int \frac{x^2 - 2x - 15 + 15}{x^2 - 2x - 15} dx \\ &= \int \frac{x^2 - 2x + 15}{x^2 - 2x - 15} dx + \int \frac{15}{x^2 - 2x - 15} dx \\ &= \int dx + \int \frac{15}{x^2 - 2x - 15} dx \\ &= \int dx - \frac{15}{8} \int \frac{1}{x + 3} dx + \frac{15}{8} \int \frac{1}{x - 5} dx \\ &= x + \frac{15}{8} [\ln(x - 5) - \ln(x + 3)] + C \\ &= x + \frac{15}{8} \ln\left(\frac{x - 5}{x + 3}\right) + C \end{aligned}$$

1 – correct expansion or equivalent merit

1 – correct integral

1 – correct answer

(b) Evaluate $\int_0^{\frac{\pi}{4}} \sec^4 x \tan^2 x \, dx$.

3

$$\begin{aligned}
 \int_0^{\frac{\pi}{4}} \sec^4 x \tan^2 x \, dx &= \int_0^{\frac{\pi}{4}} \sec^2 x \sec^2 x \tan^2 x \, dx \\
 &= \int_0^{\frac{\pi}{4}} \sec^2 x (1 + \tan^2 x) \tan^2 x \, dx \\
 &= \int_0^{\frac{\pi}{4}} \sec^2 x (\tan^2 x + \tan^4 x) \, dx \\
 &= \int_0^{\frac{\pi}{4}} \sec^2 x \tan^2 x \, dx + \int_0^{\frac{\pi}{4}} \sec^2 x \tan^4 x \, dx \\
 &= \left[\frac{\tan^3 x}{3} + \frac{\tan^5 x}{5} \right]_0^{\frac{\pi}{4}} \\
 &= \left(\frac{\tan^3 \frac{\pi}{4}}{3} + \frac{\tan^5 \frac{\pi}{4}}{5} \right) - \left(\frac{\tan^3 0}{3} + \frac{\tan^5 0}{5} \right) \\
 &= \frac{1}{3} + \frac{1}{5} = \frac{8}{15}
 \end{aligned}$$

1 – correct expansion or equivalent merit

1 – correct integral

1 – correct answer

Alternative Solution

$$\begin{aligned}
 \int_0^{\frac{\pi}{4}} \sec^4 x \tan^2 x \, dx &= \int_0^{\frac{\pi}{4}} \sec^2 x \times \sec^2 x \times \tan^2 x \, dx \\
 &= \int_0^{\frac{\pi}{4}} (1 + \tan^2 x) \times \tan^2 x \times \sec^2 x \, dx
 \end{aligned}$$

Let $u = \tan x$ when $x = \frac{\pi}{4}$ $u = 1$

$du = \sec^2 x \, dx$ when $x = 0$ $u = 0$

$$\begin{aligned}
 \int_0^1 (1 + u^2) \times u^2 \, du &= \left[\frac{u^3}{3} + \frac{u^5}{5} \right]_0^1 \\
 &= \frac{8}{15}
 \end{aligned}$$

(c) Use proof by contrapositive to prove that:

3

If $a^2 - 2a + 7$ is even then a is odd.

Contrapositive is:

If a is not odd then $a^2 - 2a + 7$ is not even.

Which is equivalent to

If a is even then $a^2 - 2a + 7$ is odd.

Let $a = 2k$ where k is an integer.

Then :

$$\begin{aligned} a^2 - 2a + 7 &= (2k)^2 - 2(2k) + 7 \\ &= 2(2k^2 - 2k + 3) + 1 \end{aligned}$$

$2k^2 - 2k + 3$ is an integer as k is an integer

$$\therefore 2(2k^2 - 2k + 3) + 1 \text{ is odd}$$

Thus If a is even then $a^2 - 2a + 7$ is odd.

Thus If $a^2 - 2a + 7$ is even then a is odd.

1 – correct contrapositive statement

1 – defining k as an integer or equivalent merit

1 – qualifying $2k^2 - 2k + 3$ as an integer and
 $2(2k^2 - 2k + 3) + 1$ as odd

(d) Solve $|e^{i\theta} + \sqrt{2}| = 1$.

3

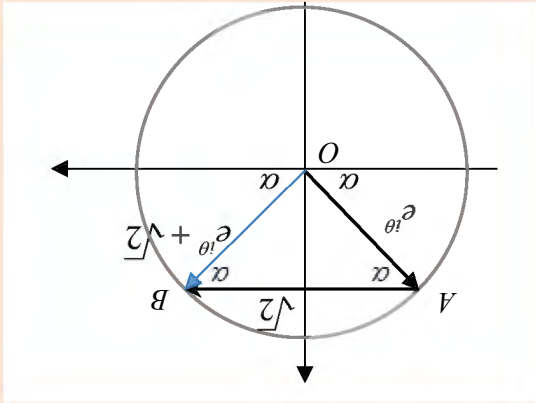
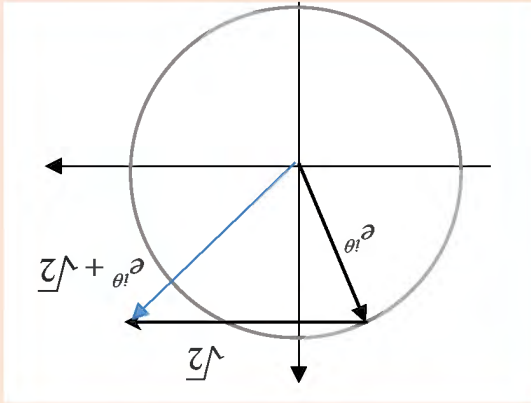
Algebraic Solution

$$\begin{aligned} |e^{i\theta} + \sqrt{2}| &= 1 \\ |\cos \theta + i \sin \theta + \sqrt{2}| &= 1 \\ (\cos \theta + \sqrt{2})^2 + (\sin \theta)^2 &= 1 \\ \cos^2 \theta + 2\sqrt{2} \cos \theta + 2 + \sin^2 \theta &= 1 \\ 2\sqrt{2} \cos \theta + 3 &= 1 \\ 2\sqrt{2} \cos \theta &= -2 \\ \cos \theta &= -\frac{1}{\sqrt{2}} \\ \theta &= -\frac{3\pi}{4} \end{aligned}$$

or $\theta = -\frac{3\pi}{4}$ for $-\pi < \theta \leq \pi$

Geometric Solution

$e^{i\theta}$ is a vector with endpoint on the unit circle. And $\sqrt{2}$ is a horizontal vector of length $\sqrt{2}$



But $|e^{i\theta} + \sqrt{2}| = 1$ and so $e^{i\theta} + \sqrt{2}$ must have its endpoint on the unit circle.
 Triangle OAB is isosceles right-angle triangle. $\alpha = \frac{\pi}{4}$ and so $\theta = -\frac{3\pi}{4}$
 A could also be in the third quadrant and so $\theta = -\frac{3\pi}{4}$

- 1 – correct calculation of magnitude
- 2 – correct answer for both angles

(e) Use the substitution $x = 2 \sin \theta$ to find $\int \frac{dx}{(4-x^2)^{\frac{3}{2}}}$.

3

$$\begin{aligned} \int \frac{dx}{(4-x^2)^{\frac{3}{2}}} &= \int \frac{2 \cos \theta}{\sqrt{(4-4 \sin^2 \theta)^3}} d\theta \\ &= \int \frac{2 \cos \theta}{8 \cos^3 \theta} d\theta \\ &= \frac{1}{4} \int \frac{1}{\cos^2 \theta} d\theta \\ &= \frac{1}{4} \int \sec^2 \theta d\theta \\ &= \frac{1}{4} \tan \theta + C \\ &= \frac{x}{4\sqrt{4-x^2}} + C \end{aligned}$$

$$\begin{aligned} x &= 2 \sin \theta \\ dx &= 2 \cos \theta d\theta \end{aligned}$$

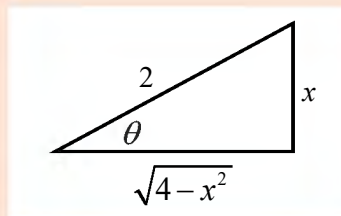
$$\frac{x}{2} = \sin \theta$$

$$\tan \theta = \frac{x}{\sqrt{4-x^2}}$$

1 – correct use of substitution

1 – correct integral in terms of θ

1 – correct integral in terms of x



End of Question 12

Question 13 (14 marks) Use a SEPARATE writing booklet

(a) Find $\int \cos^{-1} x \, dx$

3

$$I = \int \cos^{-1} x \, dx$$

$$u = \cos^{-1} x \quad v' = 1$$

$$u' = \frac{-1}{\sqrt{1-x^2}} \quad v = x$$

$$I = x \cos^{-1} x - \int x \left(\frac{-1}{\sqrt{1-x^2}} \right) dx$$

$$= x \cos^{-1} x - \frac{1}{2} \int \left(\frac{-2x}{\sqrt{1-x^2}} \right) dx$$

$$= x \cos^{-1} x - \frac{1}{2} \int \left(-2x(1-x^2)^{-\frac{1}{2}} \right) dx$$

$$= x \cos^{-1} x - \frac{1}{2} \frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$= x \cos^{-1} x - \sqrt{1-x^2} + C$$

1 – application of partial integration

1 – working towards the final solution

1 – final answer

(b) Use the substitution $t = \tan \frac{x}{2}$ or otherwise to evaluate:

3

$$\int_0^{\frac{\pi}{2}} \frac{1}{3 \sin x - 4 \cos x + 5} dx.$$

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{1}{3 \sin x - 4 \cos x + 5} dx &= \int_0^1 \frac{1}{3 \left(\frac{2t}{1+t^2} \right) - 4 \left(\frac{1-t^2}{1+t^2} \right) + 5} \times \frac{2}{1+t^2} dt \\ &= \int_0^1 \frac{2}{6t - 4(1-t^2) + 5(1+t^2)} dt \\ &= \int_0^1 \frac{2}{6t + 1 + 9t^2} dt \\ &= \frac{2}{9} \int_0^1 \frac{1}{t^2 + \frac{2}{3}t + \frac{1}{9}} dt \\ &= \frac{2}{9} \int_0^1 \frac{1}{\left(t + \frac{1}{3}\right)^2} dt \\ &= \frac{2}{9} \int_0^1 \left(t + \frac{1}{3}\right)^{-2} dt \\ &= \frac{2}{9} \left[-\left(t + \frac{1}{3}\right)^{-1} \right]_0^1 \\ &= -\frac{2}{9} \left[\left(1 + \frac{1}{3}\right)^{-1} - \left(0 + \frac{1}{3}\right)^{-1} \right] \\ &= -\frac{2}{9} \left[\frac{3}{4} - 3 \right] \\ &= -\frac{2}{9} \left[-\frac{9}{4} \right] \\ &= \frac{1}{2} \end{aligned}$$

1 – correct use of t substitution and simplification

1 – Correct Integration

1 – final answer

(c) The line l_1 passes through the points $A(2, -1, 4)$ and $B(4, -3, 2)$.

(i) Show that a vector equation of line l_1 is $\vec{r} = \lambda \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$.

2

$$\vec{r} = k\vec{AB} + \vec{OA}$$

$$\vec{OA} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$\vec{AB} = \begin{bmatrix} 4-2 \\ -3-(-1) \\ 2-4 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$\vec{r} = k(2) \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

$$= \lambda \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix} + \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$$

1 – finding vector AB

1 – for establishing correct equation and relationships, using correct vector equations and notations.

(ii) The equation of line l_2 is given by $\frac{x+3}{k} = \frac{4-y}{6} = z-1$.

3

Find the value of k for which l_1 and l_2 intersect.

Line 1

$$x = 2 + \lambda \quad \text{--- } I$$

$$y = -1 - \lambda \quad \text{--- } II$$

$$z = 4 - \lambda \quad \text{--- } III$$

Line 2

$$\frac{x+3}{k} = \frac{4-y}{6} \quad \text{--- } A$$

$$\frac{4-y}{6} = z-1 \quad \text{--- } B$$

Substitute Equations II and III into equation B :

$$\frac{4 - (-1 - \lambda)}{6} = 4 - \lambda - 1$$

$$5 + \lambda = 18 - 6\lambda$$

$$\lambda = \frac{13}{7}$$

Substitute equations I and II into equation A :

$$\frac{2 + \lambda + 3}{k} = 4 - \lambda - 1$$

$$5 + \lambda = k(3 - \lambda)$$

$$k = \frac{5 + \lambda}{3 - \lambda}$$

Substituting the value for λ

$$k = \frac{5 + \frac{13}{7}}{3 - \frac{13}{7}}$$

$$= 6$$

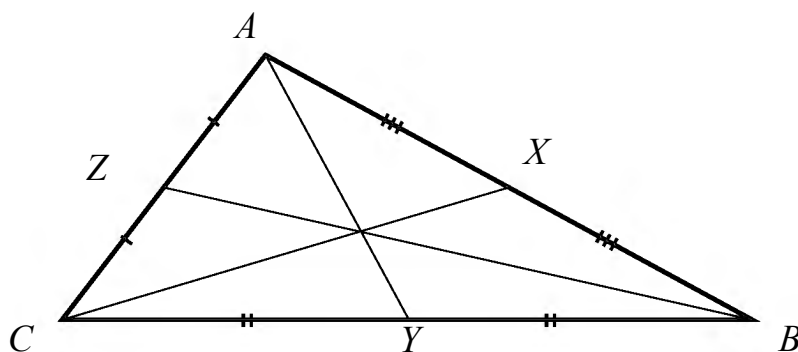
1 – Establishing relationships for line 2

1 – equating the coordinates of the two lines and calculating lambda

1 – calculation of k .

(d) The diagram shows a triangle ABC .

The points X , Y and Z bisect the intervals AB , BC and CA respectively.



Show that $\overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} = 0$

3

$$\overrightarrow{CA} + \overrightarrow{AB} + \overrightarrow{BC} = 0 \quad I$$

$$2 \times \overrightarrow{CZ} + 2 \times \overrightarrow{AX} + 2 \times \overrightarrow{BY} = 0$$

$$2 \times (\overrightarrow{CZ} + \overrightarrow{AX} + \overrightarrow{BY}) = 0$$

$$(\overrightarrow{CZ} + \overrightarrow{AX} + \overrightarrow{BY}) = 0 \quad II$$

$$\overrightarrow{AY} = \overrightarrow{CY} - \overrightarrow{CA}$$

$$\overrightarrow{BZ} = \overrightarrow{AZ} - \overrightarrow{AB}$$

$$\overrightarrow{CX} = \overrightarrow{BX} - \overrightarrow{BC}$$

By adding

$$\begin{aligned} \overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} &= (\overrightarrow{CY} - \overrightarrow{CA}) + (\overrightarrow{AZ} - \overrightarrow{AB}) + (\overrightarrow{BX} - \overrightarrow{BC}) \\ &= (\overrightarrow{CY} + \overrightarrow{AZ} + \overrightarrow{BX}) - (\overrightarrow{CA} + \overrightarrow{AB} + \overrightarrow{BC}) \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

Alternative Solution:

$$\overrightarrow{AY} = \overrightarrow{AC} + \frac{1}{2} \overrightarrow{CB}$$

$$\overrightarrow{BZ} = \overrightarrow{BA} + \frac{1}{2} \overrightarrow{AC}$$

$$\overrightarrow{CX} = \overrightarrow{CB} + \frac{1}{2} \overrightarrow{BA}$$

$$\begin{aligned} \overrightarrow{AY} + \overrightarrow{BZ} + \overrightarrow{CX} &= \overrightarrow{AC} + \frac{1}{2} \overrightarrow{CB} + \overrightarrow{BA} + \frac{1}{2} \overrightarrow{AC} + \overrightarrow{CB} + \frac{1}{2} \overrightarrow{BA} \\ &= \frac{3}{2} (\overrightarrow{CB} + \overrightarrow{BA} + \overrightarrow{AC}) \\ &= \frac{3}{2} (0) \\ &= 0 \end{aligned}$$

1 – Establishing base relations between required vectors and sides of the triangle which will lead to a solution

1 – Developing reasonable steps to lead to the answer

1 – complete solution.

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) If $x \leq 1$ and $y \geq 1$ show that $x + y \geq 1 + xy$.

2

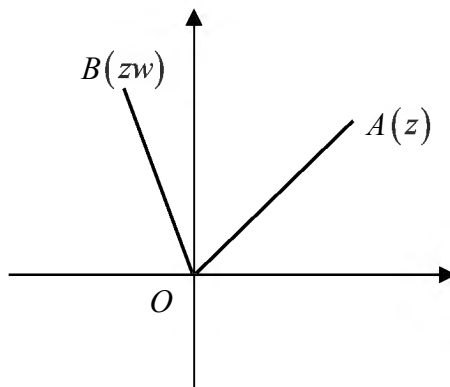
$x \leq 1$ $x - 1 \leq 0 \quad I$ $y \geq 1$ $y - 1 \geq 0 \quad II$	1 - establishing the relationships 2 - full solution
$(y-1)(x-1) \leq 0 \quad \text{From } I \text{ and } II$ $xy - x - y + 1 \leq 0$ $xy + 1 \leq x + y$ $x + y \geq xy + 1$	

- (b) Simplify fully $\frac{1}{1+\omega} + \frac{1}{1+\omega^2}$, given that ω is a non-real cube root of unity.

2

$z^3 = 1$ $z^3 - 1 = 0$ $(z-1)(1+z+z^2) = 0$ $1 + \omega + \omega^2 = 0$	1 - establishing the relationships 2 - full solution
$\frac{1}{1+\omega} + \frac{1}{1+\omega^2} = \frac{1+\omega^2+1+\omega}{(1+\omega)(1+\omega^2)}$ $= \frac{1+\omega+\omega^2+1}{1+\omega+\omega^2+\omega^3}$ $= \frac{0+1}{0+\omega^3} \quad \text{Since } 1+\omega+\omega^2 = 0$ $= \frac{1}{\omega^3} = 1 \quad \text{Since } \omega^3 = 1$	

- (c) As shown on the Argand diagram below, the complex numbers z and zw are represented by the points A and B respectively.



Given $z = re^{i\theta}$ and $w = e^{i\frac{\pi}{3}}$, where $r > 0$,

- (i) Explain why OAB is an equilateral triangle.

2

- Multiplying by z by $e^{i\frac{\pi}{3}}$ rotates the vector represented by z anticlockwise through an angle of $\frac{\pi}{3}$.

The length of z is unchanged as $|w| = 1$.

- Thus $|zw| = |z| = r$ and $\angle AOB = \frac{\pi}{3}$.

Any isosceles triangle with the angle between the equal sides being $\frac{\pi}{3}$ is an equilateral triangle.

1 – stating reason for angle being 60° as well as length of sides

2 – full solution

- (ii) Write the complex number $z - zw$ in exponential form in terms of r and θ .

2

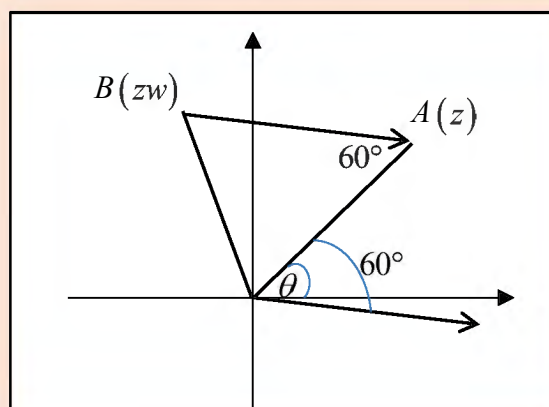
$$z - zw = \overrightarrow{BA}$$

$$\arg(z - zw) = \frac{\pi}{3} - \theta$$

$$|BA| = |OB| = |OA| = r \quad \text{Equilateral triangle}$$

$$|z - zw| = r$$

$$z - zw = re^{i(\frac{\pi}{3} - \theta)}$$



1 – answer in exponential form $re^{i\theta}$ with correct modulus;

1 – correct angle $(\frac{\pi}{3} - \theta)$

(d) (i) Use a suitable substitution to show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

1

$$RHS = \int_0^a f(a-x) dx$$

$$= \int_a^0 -f(u) du$$

$$= -\int_0^a -f(u) du$$

$$= \int_0^a f(x) dx$$

$$= LHS$$

Let $u = a - x$

$$du = -dx$$

When $x = a$ $u = 0$

$x = 0$ $u = a$

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x + \sin x} dx$.

3

$$\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x + \sin x} dx = \int_0^{\frac{\pi}{4}} \frac{\sin\left(\frac{\pi}{4} - x\right)}{\cos\left(\frac{\pi}{4} - x\right) + \sin\left(\frac{\pi}{4} - x\right)} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin \frac{\pi}{4} \cos x - \cos \frac{\pi}{4} \sin x}{\cos \frac{\pi}{4} \cos x + \sin \frac{\pi}{4} \sin x + \sin \frac{\pi}{4} \cos x - \cos \frac{\pi}{4} \sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{\cos x + \sin x + \cos x - \sin x} dx$$

$$= \int_0^{\frac{\pi}{4}} \frac{\cos x - \sin x}{2 \cos x} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \tan x) dx$$

$$= \frac{1}{2} \left[x + \ln |\cos x| \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{4} + \ln \frac{1}{\sqrt{2}} \right) - (0 + \ln 1) \right]$$

$$= \frac{1}{2} \left(\frac{\pi}{4} + \ln \frac{1}{\sqrt{2}} \right)$$

1 – Simplifying the integral

1 – Integration

1 – Final answer

- (e) A recurrence relation is defined by $u_{n+1} = \frac{3u_n - 1}{4u_n - 1}$ and $u_1 = 1$.

3

Use Mathematical Induction to prove that $u_n = \frac{n}{2n-1}$ for $n \geq 1$.

Let $n = 1$

$$LHS = u_n = 1$$

$$RHS = \frac{(1)}{2(1)-1} = \frac{1}{1} = 1$$

True for $n = 1$

Assume true for $n = k$, where k is a positive integer.

$$u_k = \frac{k}{2k-1}$$

Prove true for $n = k+1$ if true $n = k$.

$$\text{Aim: Prove that: } u_{k+1} = \frac{k+1}{2(k+1)-1}$$

$$u_{k+1} = \frac{k+1}{2k+1}$$

$$LHS = u_{n+1}$$

$$= \frac{3u_k - 1}{4u_k - 1}$$

$$= \frac{3\left(\frac{k}{2k-1}\right) - 1}{4\left(\frac{k}{2k-1}\right) - 1}$$

$$= \frac{3k - (2k-1)}{4k - (2k-1)}$$

$$= \frac{k+1}{2k+1}$$

$$= RHS$$

By Mathematical induction $u_n = \frac{n}{2n-1}$ is true for all integers, $n \geq 1$.

Question 15 (14 marks) Use a SEPARATE writing booklet

- (a) (i) Show that $a^2 + 9b^2 \geq 6ab$, where a and b are real numbers.

1

$$\begin{aligned}(a - 3b)^2 &\geq 0 \\ a^2 - 6ab + 9b^2 &\geq 0 \\ a^2 + 9b^2 &\geq 6ab\end{aligned}$$

1 – correct setting out and result

- (ii) Hence, or otherwise, show that $a^2 + 5b^2 + 9c^2 \geq 3(ab + bc + ac)$.

2

$$\begin{aligned}a^2 + 9b^2 &\geq 6ab & I \\ b^2 + 9c^2 &\geq 6bc & II \\ a^2 + 9c^2 &\geq 6ac & III \\ \text{By Adding } I, II \text{ and } III \\ a^2 + 9b^2 + b^2 + 9c^2 + a^2 + 9c^2 &\geq 6ab + 6bc + 6ac \\ 2a^2 + 10b^2 + 18c^2 &\geq 6(ab + bc + ac) \\ a^2 + 5b^2 + 9c^2 &\geq 3(ab + bc + ac)\end{aligned}$$

1 – setting up the 3 inequalities or equivalent merit

1 – correct setting and result

- (iii) Hence if $a > b > c > 0$, show that $a^2 + 5b^2 + 9c^2 > 9bc$.

2

$$\begin{aligned}a &> b > c > 0 \\ ab &> b^2 > bc > 0 \\ ab &> bc & I\end{aligned}$$

$$\begin{aligned}a &> b > c > 0 \\ ac &> bc > c^2 > 0 \\ ac &> bc & II\end{aligned}$$

$$\begin{aligned}a^2 + 5b^2 + 9c^2 &\geq 3(ab + bc + ac) && \text{From Part ii)} \\ &\geq 3(bc + bc + ac) && \text{From } I \\ &\geq 3(bc + bc + bc) && \text{From } II \\ &= 9bc\end{aligned}$$

$$\therefore a^2 + 5b^2 + 9c^2 \geq 9bc$$

1 – using given condition to setup the 3 inequalities or equivalent merit

1 – correct setting and result

(b) Consider the integral $I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} dx$.

4

Use integration by parts to show that $I_n = \frac{2n}{2n+1} I_{n-1}$.

$$I_n = \int_0^1 \frac{x^n}{\sqrt{1-x}} dx$$

$$u = x^n \quad v' = \frac{1}{\sqrt{1-x}} = (1-x)^{-\frac{1}{2}}$$

$$u' = nx^{n-1} \quad v = -2(1-x)^{\frac{1}{2}} = -2\sqrt{1-x}$$

$$I_n = \left[-2nx^{n-1}\sqrt{1-x} \right]_0^1 - \int_0^1 -2nx^{n-1}\sqrt{1-x} dx$$

$$I_n = \left[\left(-2n(1)^{n-1}\sqrt{1-1} \right) - \left(-2n(0)^{n-1}\sqrt{1-0} \right) \right]_0^1 + 2n \int_0^1 \frac{x^{n-1}(1-x)}{\sqrt{1-x}} dx$$

$$I_n = 0 + 2n \int_0^1 \frac{x^{n-1} - x^n}{\sqrt{1-x}} dx$$

$$I_n = 2n(I_{n-1} - I_n)$$

$$I_n = 2nI_{n-1} - 2nI_n$$

$$(2n+1)I_n = 2nI_{n-1}$$

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

1 – correct setup of parts

1 – correct algebraic manipulation

1 – correct integration

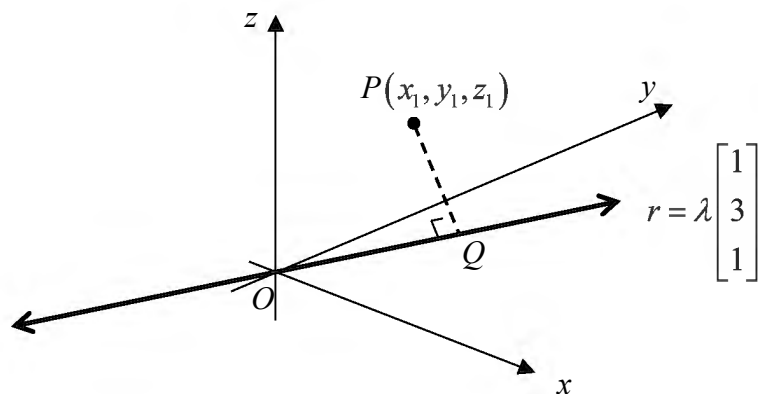
1 – correct answer

Question 15 Continues on Page 13

- (c) The diagram below shows the line with vector equation $\underline{r} = \lambda \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

The point $P(x_1, y_1, z_1)$ is any point in the three-dimensional space.

Q is the point on $\underline{r}$ such that PQ is a minimum.



- (i) Using vector projection methods show that:

2

$$\overrightarrow{OQ} = \frac{x_1 + 3y_1 + z_1}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

$\overrightarrow{OQ}$ is the projection of $\overrightarrow{OP}$ onto line $\underline{r} = \lambda \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$.

Let $\overrightarrow{OP} = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix}$ and $\underline{b} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$ which is a vector on $\underline{r}$

$$\overrightarrow{OQ} = \text{proj}_{\underline{b}} \overrightarrow{OP}$$

$$= \frac{\overrightarrow{OP} \cdot \underline{b}}{|\underline{b}|^2} \times \underline{b}$$

$$= \frac{(1)(x_1) + (3)(y_1) + (1)(z_1)}{1^2 + 3^2 + 1^2} \times \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

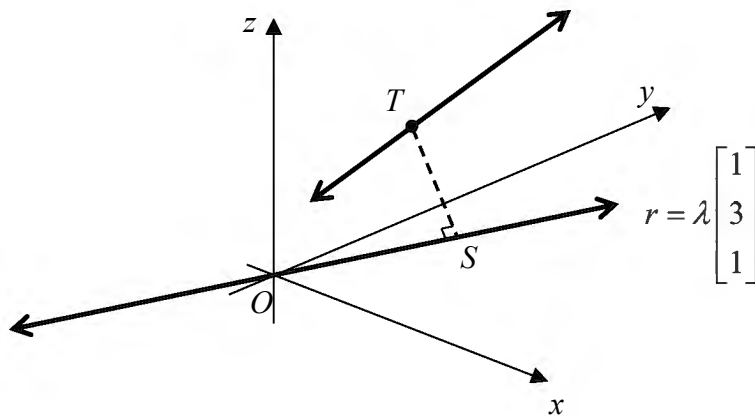
$$= \frac{x_1 + 3y_1 + z_1}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

1 – correct projection formula and application

1 – correct answer

It is given that line $\zeta = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$ does not intersect $\frac{r}{11}$.

Point T is on the line ζ and S is a point on $\frac{r}{11}$ such that TS is a minimum.



- (ii) By first expressing ζ in parametric form, find an expression for the vector $\overrightarrow{TS}$ in terms of t .

3

$$\zeta = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix} \text{ in parametric form is } \zeta = \begin{bmatrix} 1-t \\ 4+t \\ 5t \end{bmatrix}$$

If T is any point on $\zeta = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 1 \\ 5 \end{bmatrix}$ then the vector $\overrightarrow{OT}$ in terms of t is $\begin{bmatrix} 1-t \\ 4+t \\ 5t \end{bmatrix}$.

By substituting the parametric equation into the result in part i)

$$\overrightarrow{OS} = \frac{(1-t) + 3(4+t) + (5t)}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\overrightarrow{OS} = \frac{13+7t}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\overrightarrow{TS} = \overrightarrow{OS} - \overrightarrow{OT} = \frac{13+7t}{11} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} - \begin{bmatrix} 1-t \\ 4+t \\ 5t \end{bmatrix}$$

$$= \frac{1}{11} \begin{bmatrix} (13+7t) - 11(1-t) \\ 3(13+7t) - 11(4+t) \\ (13+7t) - 11(5t) \end{bmatrix}$$

$$= \frac{1}{11} \begin{bmatrix} 2+18t \\ -5+10t \\ 13-48t \end{bmatrix}$$

1 – correct parametric representation of T as a point on ζ

1 – correct calculation of $\overrightarrow{TS}$

1 – correct answer

Question 16 (15 marks) Use a SEPARATE writing booklet

(a) (i) Prove that:

2

$$\frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1} = i \tan \frac{\theta}{2}.$$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\begin{aligned} LHS &= \frac{\cos \theta + i \sin \theta - 1}{\cos \theta + i \sin \theta + 1} \\ &= \frac{\frac{1-t^2}{1+t^2} + i \frac{2t}{1+t^2} - 1}{\frac{1-t^2}{1+t^2} + i \frac{2t}{1+t^2} + 1} \\ &= \frac{1-t^2+2ti-1-t^2}{1-t^2+2ti+1+t^2} \\ &= \frac{2ti-2t^2}{2+2ti} \\ &= \frac{t(i-t)}{1+ti} \\ &= \frac{ti\left(1-\frac{t}{i}\right)}{1+ti} \\ &= \frac{ti(1+ti)}{1+ti} \\ &= it \\ &= i \tan \frac{\theta}{2} \end{aligned}$$

(ii) Find the roots of the equation $w^5 = 1$.

2

Express your answers in the form $(\cos \theta + i \sin \theta)$ where $-\pi < \theta \leq \pi$.

$$\begin{aligned} (\cos \theta + i \sin \theta)^5 &= 1 \\ (\cos 5\theta + i \sin 5\theta) &= (\cos 2k\pi + i \sin 2k\pi) \quad \text{Where } k \text{ is an integer.} \\ 5\theta &= 2k\pi \\ \theta &= \frac{2k\pi}{5} \\ w &= \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5} \quad \text{Where } k = 0, \pm 1, \pm 2 \end{aligned}$$

(iii) Using parts (i) and (ii) find the roots of the equation:

3

$$\left(\frac{2+z}{2-z}\right)^5 = 1$$

Since $w^5 = 1$

$$\frac{2+z}{2-z} = w$$

$$2+z = w(2-z)$$

$$2+z = 2w - zw$$

$$zw + z = 2w - 2$$

$$z(w+1) = 2(w-1)$$

$$z = \frac{2(w-1)}{w+1}$$

Since $w^5 = 1$ and from part (ii)

$$w = \cos \theta + i \sin \theta$$

$$\theta = \frac{2k\pi}{5} \quad \text{Where } k = 0, \pm 1, \pm 2$$

And from Part (i)

$$\begin{aligned} z &= \frac{2(\cos \theta + i \sin \theta - 1)}{\cos \theta + i \sin \theta + 1} \\ &= 2i \tan \frac{\theta}{2} \end{aligned}$$

$$\text{For } \theta = \frac{2k\pi}{5} \quad \text{Where } k = 0, \pm 1, \pm 2$$

$$z = 2i \tan\left(\frac{-2\pi}{5}\right), \quad 2i \tan\left(\frac{-\pi}{5}\right), \quad 2i \tan(0), \quad 2i \tan\left(\frac{\pi}{5}\right), \quad 2i \tan\left(\frac{2\pi}{5}\right)$$

- (b) (i) Show that $\frac{1}{(k-1)k(k+1)} \geq \frac{1}{k^3}$ where k is a positive integer greater than 1. 2
- (ii) Given that: 3

$$S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} \text{ where } n \text{ is an integer and } n \geq 3.$$

$$\frac{1}{(k-1)k(k+1)} = \frac{A}{(k-1)} + \frac{B}{k} + \frac{C}{(k+1)}$$

$$1 = Ak(k+1) + B(k-1)(k+1) + C(k-1)k$$

$k = 0$	$k = 1$	$k = 1$
$1 = B(-1)(1)$	$1 = A(1+1)$	$1 = +C(-1-1)(-1)$
$B = -1$	$A = \frac{1}{2}$	$C = \frac{1}{2}$

$$\frac{1}{(k-1)k(k+1)} = \frac{1}{2(k-1)} - \frac{1}{k} + \frac{1}{2(k+1)}$$

$$\frac{1}{k^3} \leq \frac{1}{2(k-1)} - \frac{1}{k} + \frac{1}{2(k+1)}$$

$$k = 3 \quad \frac{1}{3^3} \leq \frac{1}{2 \times 2} - \frac{1}{3} + \frac{1}{2 \times 4}$$

$$k = 4 \quad \frac{1}{4^3} \leq \frac{1}{2 \times 3} - \frac{1}{4} + \frac{1}{2 \times 5}$$

$$k = 5 \quad \frac{1}{5^3} \leq \frac{1}{2 \times 4} - \frac{1}{5} + \frac{1}{2 \times 6}$$

$$k = 6 \quad \frac{1}{6^3} \leq \frac{1}{2 \times 5} - \frac{1}{6} + \frac{1}{2 \times 7}$$

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$$k = n-2 \quad \frac{1}{(n-2)^3} \leq \frac{1}{2 \times (n-3)} - \frac{1}{(n-2)} + \frac{1}{2 \times (n-1)}$$

$$k = n-1 \quad \frac{1}{(n-1)^3} \leq \frac{1}{2 \times (n-2)} - \frac{1}{(n-1)} + \frac{1}{2 \times n}$$

$$k = n \quad \frac{1}{(n-1)^3} \leq \frac{1}{2 \times (n-1)} - \frac{1}{n} + \frac{1}{2 \times (n+1)}$$

Using part (i) or otherwise, prove that $S_n < \frac{1}{12}$.

By adding and cancelling

$$S_n \leq \frac{1}{4} - \frac{1}{3} + \frac{1}{6} - \frac{1}{2n} - \frac{1}{n} + \frac{1}{2n+2}$$

$$S_n \leq \frac{1}{12} - \frac{1}{2n} + \frac{1}{2n+2}$$

$$S_n \leq \frac{1}{12} + \frac{-n-1+n}{2n(n+1)}$$

$$S_n \leq \frac{1}{12} + \frac{-1}{2n(n+1)}$$

$$S_n \leq \frac{1}{12} \quad \text{Since } \frac{-1}{2n(n+1)} < 0$$

(c) Find $\int \frac{\ln x - 1}{[x + \ln x]^2} dx$.

$$\begin{aligned} I &= \int \frac{\ln x - 1}{[x + \ln x]^2} dx \\ &= \int \frac{x + \ln x - x - 1}{[x + \ln x]^2} dx \\ &= \int \frac{x + \ln x}{[x + \ln x]^2} dx - \int \frac{x + 1}{[x + \ln x]^2} dx \\ &= \int \frac{1}{x + \ln x} dx - \int \frac{x(1 + \frac{1}{x})}{[x + \ln x]^2} dx \\ &= \int \frac{1}{x + \ln x} dx - J \end{aligned}$$

$$\text{Let } J = \int \frac{x(1 + \frac{1}{x})}{[x + \ln x]^2} dx$$

By using integration by parts on J

$$u = x \quad v' = \frac{(1 + \frac{1}{x})}{[x + \ln x]^2}$$

$$u' = 1 \quad v = \frac{-1}{x + \ln x} \quad \text{by Reverse Chain Rule}$$

$$\begin{aligned} J &= x \left(\frac{-1}{x + \ln x} \right) - \int \frac{-1}{x + \ln x} dx \\ &= \frac{-x}{x + \ln x} + \int \frac{1}{x + \ln x} dx \end{aligned}$$

$$\begin{aligned} &= \int \frac{1}{x + \ln x} dx - \left(\frac{-x}{x + \ln x} + \int \frac{1}{x + \ln x} dx \right) \\ &= \frac{x}{x + \ln x} + C \end{aligned}$$

Alternative Solutions

14 (a)

Alternative 1:

Assume $x + y < 1 + xy$

Then $xy - x - y + 1 > 0$

$$x(y-1) - 1(y-1) > 0$$

$$(x-1)(y-1) > 0$$

But $x-1 \leq 0$ ($x \leq 1$)

and $y-1 \geq 0$ ($y \geq 1$)

Therefore $(x-1)(y-1) \leq 0$

Therefore, by contradiction, $x + y \geq 1 + xy$

Alternative 2:

$y \geq 1$ and $1 - x \geq 0$ (Given)

Therefore $y(1-x) \geq (1-x)$

$$y - xy \geq 1 - x$$

$$x + y \geq xy + 1$$

14 (b)**Alternative 1:**

$$\begin{aligned}
\frac{1}{1+\omega} + \frac{1}{1+\omega^2} &= \frac{1+\omega^2+1+\omega}{(1+\omega)(1+\omega^2)} \\
&= \frac{2+\omega+\omega^2}{1+\omega+\omega^2+\omega^3} \\
&= \frac{2+\omega+\omega^2}{1+\omega+\omega^2+1} && \text{Since } \omega^3 = 1 \\
&= \frac{2+\omega+\omega^2}{2+\omega+\omega^2} \\
&= 1
\end{aligned}$$

Alternative 2:

$$\begin{aligned}
\omega^3 &= 1 \\
\omega &= -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i && \text{since } \omega \text{ is complex cube root of unity.} \\
\frac{1}{1+\omega} + \frac{1}{1+\omega^2} &= \frac{1}{1-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i} + \frac{1}{1+\left(-\frac{1}{2} \pm \frac{\sqrt{3}}{2}i\right)^2} \\
&= \frac{1}{\frac{1}{2} \pm \frac{\sqrt{3}}{2}i} + \frac{1}{1+\frac{1}{4} - \frac{3}{4} \mp \frac{\sqrt{3}}{2}i} \\
&= \frac{2}{1 \pm \sqrt{3}i} + \frac{2}{1 \mp \sqrt{3}i} \\
&= \frac{2(1-\sqrt{3}i) + 2(1+\sqrt{3}i)}{(1+\sqrt{3}i)(1-\sqrt{3}i)} \quad \text{or} \quad \frac{2(1+\sqrt{3}i) + 2(1-\sqrt{3}i)}{(1-\sqrt{3}i)(1+\sqrt{3}i)} \\
&= \frac{4}{1+3} \\
&= 1
\end{aligned}$$

14 (d) (i)

$$\begin{aligned} LHS &= \int_0^a f(x) dx \\ &= \int_a^0 -f(a-u) du \\ &= -\int_0^a -f(a-u) du \\ &= \int_0^a f(a-u) du \\ &= \int_0^a f(a-x) dx \\ &= RHS \end{aligned}$$

$$\text{Let } x = a - u$$

$$dx = -du$$

$$\text{When } x = a \quad u = 0$$

$$x = 0 \quad u = a$$

End of paper